Review Notes: Analysis in $\mathbb{R}^n$

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1 Differentiation in $\mathbb{R}$

1.1 Lecture 1 — Real Functions

Definition 1.1: Differentiability

Let $f : [a, b] \to \mathbb{R}$ and $x \in [a, b]$. We say f is *differentiable at* x if

$$\lim_{t \to x} \frac{f(t) - f(x)}{t - x}$$

exists. If this limit exists, it is called the *derivative* of f at x, denoted $f'(x)$.

Remark.

> Recall: $\lim_{t \to x} g(t) = y$ exists means: $\forall \varepsilon > 0$, $\exists \delta > 0$ such that $\forall t \in [a, b]$, if $0 < |t - x| < \delta$ then $|g(t) - y| < \varepsilon$.

Extensions

(1) Complex-valued functions. $f : [a, b] \to \mathbb{C}$, write $f = f_{\text{Re}} + i f_{\text{Im}}$ where $f_{\text{Re}}, f_{\text{Im}} : [a, b] \to \mathbb{R}$. If f is differentiable at x, then

$$f'(x) = f'_{\text{Re}}(x) + i\, f'_{\text{Im}}(x).$$

(2) Vector-valued functions. $f : [a, b] \to \mathbb{R}^n$, write $f = (f_1, \ldots, f_n)$ with $f_i : [a, b] \to \mathbb{R}$ for all $1 \le i \le n$. The derivative is defined the same way:

$$f'(x) = \lim_{t \to x} \frac{f(t) - f(x)}{t - x}$$

where $f(x)$ and $f(t)$ are vectors. By Rudin Theorem 4.10, the limit exists if and only if the limit of each component exists, so

$$f'(x) = \big(f'_1(x),\, f'_2(x),\, \ldots,\, f'_n(x)\big).$$

Theorem 1.2: Differentiability implies continuity

If $f : [a, b] \to \mathbb{R}$ is differentiable at $x \in [a, b]$, then f is continuous at x.

Proof. For any $t \neq x$,

$$f(t) - f(x) = \frac{f(t) - f(x)}{t - x} \cdot (t - x).$$

As $t \to x$ the first factor converges to $f'(x)$ (well-defined by differentiability) and $(t - x) \to 0$, so $f(t) - f(x) \to 0$.

Exercise. Complete the argument using the ε-δ definition: show that $\forall \varepsilon > 0\ \exists \delta > 0$ such that $|f(t) - f(x)| < \varepsilon$ whenever $|t - x| < \delta$. (Review the definition and limit arithmetics.) $\square$

1.2 Differentiation Rules

Proposition 1.3

(1) Let $f, g : [a, b] \to \mathbb{R}$ be differentiable at x. Then $f + g$, fg, and f/g (when $g \neq 0$) are also differentiable at x, with:

$$\begin{aligned}
&(a) \quad (f+g)'(x) = f'(x) + g'(x), \\
&(b) \quad (fg)'(x) = f'(x)g(x) + f(x)g'(x), \\
&(c) \quad \left(\tfrac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}. \quad \textit{(prove this!)}
\end{aligned}$$

(2) Chain rule. Let $f : [a, b] \to \mathbb{R}$, $g : [c, d] \to \mathbb{R}$ with $f([a, b]) \subseteq [c, d]$. If f is differentiable at x and g is differentiable at $f(x)$, then $g \circ f$ is differentiable at x and

$$(g \circ f)'(x) = g'(f(x)) \cdot f'(x).$$

Remark.

> **Thought of proof for (2).**
>
> $$\frac{g(f(t)) - g(f(x))}{t - x} = \frac{g(f(t)) - g(f(x))}{f(t) - f(x)} \cdot \frac{f(t) - f(x)}{t - x}.$$
>
> As $t \to x$: $\dfrac{f(t) - f(x)}{t - x} \to f'(x)$. However, the denominator $f(t) - f(x)$ can equal 0, so making this rigorous requires extra care.
>
> Everything so far applies to $\mathbb{R}^n$, $\mathbb{C}$, vector-valued f's. For part (b), replace the product with the dot product.

Remark.

> Derivatives are connected to: local max/min, monotonicity, and range of values.

1.3 Local Extrema

Definition 1.4: Local maximum / minimum

Let (X, d) be a metric space and $f : X \to \mathbb{R}$. f has a *local maximum* at $x \in X$ if there exists an open ball $B \ni x$ such that $\forall y \in B : \ f(y) \leq f(x)$. (A *local minimum* satisfies $f(y) \geq f(x)$.)

Theorem 1.5: Interior extremum theorem

Let $f : [a, b] \to \mathbb{R}$ and suppose $x \in (a, b)$ is a local maximum or a local minimum. If $f'(x)$ exists, then $f'(x) = 0$.

Proof. Let x be a local maximum, so $\exists \delta > 0$ with $f(y) \leq f(x)$ for all $y \in B(x, \delta)$. Then for any $0 < |t - x| < \delta$:

$$t < x : \quad \frac{f(t) - f(x)}{t - x} \geq 0 \qquad \text{and} \qquad t > x : \quad \frac{f(t) - f(x)}{t - x} \leq 0.$$

Taking $t \to x$ from both sides, $f'(x) \geq 0$ and $f'(x) \leq 0$, so $f'(x) = 0$. □

1.4 Lecture 2 — Mean Value Theorem

Theorem 1.6: Generalized Mean Value Theorem (GMVT)

Hypotheses. Let $f, g : [a, b] \to \mathbb{R}$, both continuous on $[a, b]$ and differentiable on (a, b).
Conclusion. Then $\exists\, \lambda \in (a, b)$ such that

$$\big(f(b) - f(a)\big) \cdot g'(\lambda) = \big(g(b) - g(a)\big) \cdot f'(\lambda).$$

Remark.

> If $g(b) - g(a) \neq 0$ and $g'(\lambda) \neq 0$, then:
>
> $$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(\lambda)}{g'(\lambda)}.$$
>
> That is, there is a point λ where the ratio of the derivatives equals the ratio of the net changes — geometrically, the two curves have the same "slope" at λ.
>
> Taking $g(x) = x$ gives $g(b) - g(a) = b - a$ and $g'(x) = 1$, which recovers the standard MVT.

Corollary 1.7

Standard MVT. Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Then $\exists\, \lambda \in (a, b)$ such that

$$f'(\lambda) = \frac{f(b) - f(a)}{b - a}.$$

Proof. **Idea:** Build h from f and g, then find λ with $h'(\lambda) = 0$.

Define

$$h(x) = \big(f(b) - f(a)\big)g(x) - \big(g(b) - g(a)\big)f(x).$$

Then

$$h'(x) = \big(f(b) - f(a)\big)g'(x) - \big(g(b) - g(a)\big)f'(x).$$

WTS: $\exists\, \lambda \in (a, b)$ with $h'(\lambda) = 0$.

Compute the endpoint values:

$$\begin{aligned} h(a) &= f(b)g(a) - f(a)g(a) - g(b)f(a) + g(a)f(a) = f(b)g(a) - g(b)f(a), \\ h(b) &= f(b)g(b) - f(a)g(b) - f(b)g(b) + g(a)f(b) = -f(a)g(b) + f(b)g(a). \end{aligned}$$

So $h(a) = h(b)$. We seek an interior max or min.

- **Case 1:** h is constant $\Rightarrow h'(\lambda) = 0$ for all $\lambda \in (a, b)$. ✓
- **Case 2:** h is not constant, so $\exists\, x \in (a, b)$ with $h(x) \neq h(a)$. WLOG $h(x) > h(a) = h(b)$. By the Extreme Value Theorem, h attains a global maximum at some $\lambda \in [a, b]$. Since $\sup_{y \in [a,b]} h(y) \geq h(x) > h(a) = h(b)$, we have $\lambda \notin \{a, b\}$, so $\lambda \in (a, b)$. By the interior extremum theorem, $h'(\lambda) = 0$. ✓

□

Corollary 1.8

Monotonicity (Corollary of Standard MVT). If $f : [a, b] \to \mathbb{R}$ is differentiable, then:

1. If $f'(x) \geq 0$ for all $x \in (a, b)$, then f is non-decreasing.
2. If $f'(x) \leq 0$ for all $x \in (a, b)$, then f is non-increasing.
3. If $f'(x) = 0$ for all $x \in (a, b)$, then f is constant.

Proof. We prove (1); the others are analogous. Let $a < x < y < b$. By the MVT, $\exists\, \lambda \in (x, y)$ such that

$$\frac{f(y) - f(x)}{y - x} = f'(\lambda) \geq 0,$$

so $f(y) - f(x) = f'(\lambda) \cdot (y - x) \geq 0$, i.e. $f(x) \leq f(y)$. □

1.5 Intermediate Value Property for Derivatives

Remark.

> **Q:** Can any function be written as the derivative of something? That is, given f, does $g' = f$ always have a solution? By calculus: if f is continuous, then $g' = f$ does.
> **A:** No — derivatives must satisfy the **Intermediate Value Property (IVP)**.

Theorem 1.9: Intermediate Value Property for Derivatives

Let $f : [a, b] \to \mathbb{R}$ be differentiable on $[a, b]$. Suppose c is strictly between $f'(a)$ and $f'(b)$ (say $f'(a) < c < f'(b)$). Then $\exists\, \lambda \in (a, b)$ such that $f'(\lambda) = c$.

Proof. Set $g(x) = f(x) - cx$. Then

$$g'(a) = f'(a) - c < 0 \qquad \text{and} \qquad g'(b) = f'(b) - c > 0.$$

Since $g'(x) = f'(x) - c$, it suffices to find $\lambda \in (a, b)$ with $g'(\lambda) = 0$.

Let $\lambda \in [a, b]$ be a global minimum of g (which exists by EVT).

- Suppose $\lambda = a$. Then
$$g'(a) = \lim_{x \to a^+} \frac{g(x) - g(a)}{x - a} \geq 0,$$
contradicting $g'(a) < 0$.
- Suppose $\lambda = b$. Then similarly $g'(b) \leq 0$, contradicting $g'(b) > 0$.

Hence $\lambda \in (a, b)$, and since λ is an interior minimum, $g'(\lambda) = 0$. $\square$

Remark.

The function
$$f(x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$$
is *not* a derivative, since it fails the IVP (it skips all values strictly between -1 and 1).

Remark.

The MVT does *not* hold for functions $f : [a, b] \to \mathbb{R}^n$ (or $\mathbb{C}$).
Example. $f : [0, 2\pi] \to \mathbb{R}^2$, $f(t) = (\cos t, \sin t)$. If MVT held, then $\exists\, \lambda \in (0, 2\pi)$ such that
$$f'(\lambda) = \frac{f(2\pi) - f(0)}{2\pi - 0} = \mathbf{0}.$$
But $f'(t) = (-\sin t, \cos t) \neq \mathbf{0}$ for all t. (Geometrically: the circle has no tangent parallel to the secant, which is a point.)

1.6 L'Hôpital's Rule

Remark.

Technical aside on limits at endpoints. Let $h : (a, b) \to \mathbb{R}$ with $-\infty \leq a < b \leq \infty$. To prove $\lim_{x \to a} h(x) = L \in [-\infty, \infty]$ it suffices to show:

- If $-\infty \leq L < \infty$: for all $q > L$, $\exists\, c \in (a, b)$ such that $a < x < c \Rightarrow h(x) < q$.
- If $-\infty < L \leq \infty$: for all $p < L$, $\exists\, c \in (a, b)$ such that $a < x < c \Rightarrow h(x) > p$.

Theorem 1.10: L'Hôpital's Rule

Let $f, g : (a, b) \to \mathbb{R}$ be differentiable with $-\infty \leq a < b \leq \infty$. Assume $g'(x) \neq 0$ for all $x \in (a, b)$ and that $\lim_{x \to a} \dfrac{f'(x)}{g'(x)}$ exists in $\mathbb{R} \cup \{-\infty, +\infty\}$.

(i) If $\lim_{x \to a} f(x) = \lim_{x \to a} g(x) = 0$, then $\lim_{x \to a} \dfrac{f(x)}{g(x)} = \lim_{x \to a} \dfrac{f'(x)}{g'(x)}$.

(ii) If $\lim_{x \to a} g(x) = \pm\infty$, then $\lim_{x \to a} \dfrac{f(x)}{g(x)} = \lim_{x \to a} \dfrac{f'(x)}{g'(x)}$.

Proof. Let $L = \lim_{x \to a} \dfrac{f'(x)}{g'(x)}$ (exists by assumption).

First assume $-\infty \leq L \leq \infty$. It suffices to show: for all $q > L$, $\frac{f(x)}{g(x)} < q$ for all x sufficiently close to a.

Choose r with $L < r < q$. Since $\frac{f'}{g'} \to L$, there exists $c \in (a, b)$ such that

$$\frac{f'(x)}{g'(x)} < r \qquad \forall x \in (a, c).$$

For $x, y \in (a, b)$ with $a < x < y < c$, by the GMVT there exists $\lambda \in (x, y)$ such that

$$\frac{f(y) - f(x)}{g(y) - g(x)} = \frac{f'(\lambda)}{g'(\lambda)} < r.$$

Case (i): Assume $\lim_{x \to a} f(x) = \lim_{x \to a} g(x) = 0$. Fix $y \in (a, c)$. For all x with $a < x < y$:

$$\frac{f(y) - f(x)}{g(y) - g(x)} < r.$$

As $x \to a$, the left side $\to \frac{f(y)}{g(y)}$, so $\frac{f(y)}{g(y)} \leq r < q$.

Case (ii): Assume $\lim_{x \to a} g(x) = +\infty$. Fix $y \in (a, c)$. For all $x \in (a, y)$:

$$\frac{f(y) - f(x)}{g(y) - g(x)} < r.$$

Let $d \in (a, y)$ such that $\forall x \in (a, d)$: $g(x) > 0$ and $g(x) > g(y)$. Then for every $x \in (a, d)$:

$$r > \frac{f(y) - f(x)}{g(y) - g(x)} = \frac{\frac{f(y)}{g(x)} - \frac{f(x)}{g(x)}}{\frac{g(y)}{g(x)} - 1}$$

where $\frac{g(y)}{g(x)} - 1 < 0$ (since $g(x) > g(y)$). This implies

$$\frac{f(x)}{g(x)} < r\left(1 - \frac{g(y)}{g(x)}\right) + \frac{f(y)}{g(x)}.$$

As $x \to a$, the right side $\to r$. In particular,

$$\lim_{x \to a} \left[r\left(1 - \frac{g(y)}{g(x)}\right) + \frac{f(y)}{g(x)} \right] = r,$$

so $\exists\, d_2 \in (a, d)$ such that $\frac{f(x)}{g(x)} < q$ for all $x \in (a, d_2)$. $\square$

1.7 Taylor's Theorem

Definition 1.11: Higher-order derivatives

For $n \in \mathbb{N}$, the n-th derivative $f^{(n)}$ is defined inductively by applying the definition of derivative to $f^{(n-1)}$.

Remarks.

1. If $f^{(n)}$ exists at x, then $f^{(n-1)}$ must exist in a neighborhood of x.
2. If $f^{(n)}$ exists on (a, b), then $f', f'', \ldots, f^{(n-1)}$ must exist and be continuous on (a, b).

Definition 1.12: Taylor polynomial

Let $f : [a, b] \to \mathbb{R}$, $n \in \mathbb{N}$ with $f^{(n)}$ existing on (a, b), and $c \in (a, b)$. The *Taylor polynomial of degree* k of f at c is

$$P_k(x) = \sum_{i=0}^{k} \frac{f^{(i)}(c)}{i!}(x - c)^i.$$

In particular, $P_0(x) = f(c)$ and $P_1(x) = f(c) + f'(c)(x - c)$.

Theorem 1.13: Taylor's Theorem

Let $f : [a, b] \to \mathbb{R}$ with $f^{(n)}$ existing on (a, b) and $c \in (a, b)$. For any $x \in (a, b)$ with $x \neq c$, there exists λ strictly between x and c such that

$$f(x) = \underbrace{P_{n-1}(x)}_{\text{polynomial approximation}} + \underbrace{\frac{f^{(n)}(\lambda)}{n!}(x-c)^n}_{\text{error term}}.$$

Note. When $n = 1$: $f(x) = f(c) + f'(\lambda)(x - c)$, which is exactly the MVT.

Proof. **Setup.** Let $M_x \in \mathbb{R}$ be defined by

$$f(x) = P_{n-1}(x) + M_x(x-c)^n.$$

It suffices to show $\exists\, \lambda$ between x and c with $M_x = \frac{f^{(n)}(\lambda)}{n!}$.

Strategy: Define a function and show it vanishes somewhere (repeatedly apply MVT). Let

$$g(y) = f(y) - P_{n-1}(y) - M_x(y-c)^n.$$

(1) $g(x) = 0$ by definition of M_x.

(2) $g(c) = f(c) - P_{n-1}(c) = 0$ (since $P_{n-1}(c) = f(c)$).

(3) For all $k \leq n - 1$:

$$g^{(k)}(y) = f^{(k)}(y) - P_{n-1}^{(k)}(y) - M_x \cdot n(n-1)\cdots(n-k+1)(y-c)^{n-k}.$$

In particular, $g^{(k)}(c) = f^{(k)}(c) - P_{n-1}^{(k)}(c) = 0$, because the k-th derivative of P_{n-1} at c equals $f^{(k)}(c)$.

So $g(x) = 0$ and $g(c) = g'(c) = \cdots = g^{(n-1)}(c) = 0$. Apply MVT repeatedly:

$$\begin{aligned}
&\exists\, \lambda_1 \text{ between } c \text{ and } x: \ g'(\lambda_1) = 0,\\
&\exists\, \lambda_2 \text{ between } c \text{ and } \lambda_1: \ g''(\lambda_2) = 0,\\
&\qquad\vdots\\
&\exists\, \lambda_n \text{ between } c \text{ and } \lambda_{n-1}: \ g^{(n)}(\lambda_n) = 0.
\end{aligned}$$

Now, P_{n-1} is a polynomial of degree $n - 1$, so $P_{n-1}^{(n)} \equiv 0$. Thus

$$0 = g^{(n)}(\lambda_n) = f^{(n)}(\lambda_n) - M_x \cdot n!,$$

giving $M_x = \dfrac{f^{(n)}(\lambda_n)}{n!}$. Set $\lambda = \lambda_n$. $\square$

2 Differentiation in $\mathbb{R}^n$

2.1 Vector Spaces and Linear Algebra Review

Definition 2.1: Vector space, span, basis

1. A *vector space* (over $\mathbb{R}$) is a non-empty set $X \subseteq \mathbb{R}^n$ satisfying: (i) $\forall\, x, y \in X$: $x + y \in X$; and (ii) $\forall\, x \in X$, $a \in \mathbb{R}$: $ax \in X$.
2. $x_1, \ldots, x_k \in \mathbb{R}^n$. A *linear combination* is any vector of the form $c_1x_1 + \cdots + c_kx_k$ with $c_i \in \mathbb{R}$.
3. The *span* of a non-empty $S \subseteq \mathbb{R}^n$ is the set of all linear combinations of elements of S.
4. $\{x_1, \ldots, x_k\}$ is *linearly independent* if $a_1x_1 + \cdots + a_kx_k = 0 \Rightarrow a_1 = \cdots = a_k = 0$.
5. X has *dimension* r if it contains r linearly independent vectors but no set of $r + 1$ linearly independent vectors.
6. An independent set $\{x_1, \ldots, x_r\} \subseteq X$ that spans X is called a *basis* of X.
7. If $B = \{x_1, \ldots, x_r\}$ is a basis of X, then every $x \in X$ can be written uniquely as $x = a_1x_1 + \cdots + a_rx_r$ (the scalars $a_1, \ldots, a_r$ are called the *coordinates* of x w.r.t. B).

Theorem 2.2: Basis theorem

Let X be a vector space with $\dim X = n$. Then:

1. If $S \subseteq X$ with $|S| = n$, then $X = \operatorname{span} S \iff S$ is linearly independent.
2. X has a basis.
3. Every basis has exactly n elements.
4. If $m \leq n$ and $\{y_1, \ldots, y_m\} \subseteq X$ is linearly independent, then $\exists$ a basis of X containing $\{y_1, \ldots, y_m\}$.

2.2 Lecture 5 (1/14) — Linear Transformations

Throughout, X, Y, Z denote vector spaces.

Definition 2.3: Linear transformation, $L(X, Y)$

1. $A : X \to Y$ is a *linear transformation* (or *linear operator*) if for all $x_1, x_2 \in X$ and $c_1, c_2 \in \mathbb{R}$:
$$A(c_1x_1 + c_2x_2) = c_1A(x_1) + c_2A(x_2).$$
2. $L(X, Y)$ denotes the set of all linear transformations from X to Y. Write $L(X) = L(X, X)$.
3. If $A \in L(X, Y)$ and A^{-1} exists, then A is called *invertible* and A^{-1} is its inverse.
4. If $A \in L(X, Y)$ and $B \in L(Y, Z)$, their product (composition) is $BA : X \to Z$, $(BA)(x) = B(Ax)$.

Remark.

1. $BA \in L(X, Z)$. (Knowing A on a finite basis determines A everywhere.)
2. If $A \in L(X, Y)$ and A^{-1} exists, then $A^{-1} \in L(Y, X)$.
3. If $A \in L(X, Y)$ and $\{x_1, \ldots, x_n\}$ is a basis of X, then A is completely determined by the values Ax_j. Indeed, if $x = \sum c_jx_j$ then $Ax = A(\sum c_jx_j) = \sum c_jAx_j$.

Proposition 2.4

Matrix representation. Every linear transformation has a matrix representation. Let $A \in L(X, Y)$ where X has basis $B_X = \{x_1, \ldots, x_n\}$ and Y has basis $B_Y = \{y_1, \ldots, y_m\}$. If $x = \sum_{j=1}^n c_jx_j$, then $Ax = \sum_{j=1}^n c_jAx_j$. Expressing each Ax_j in terms of B_Y:
$$Ax_j = \sum_{i=1}^m a_{ij}\, y_i, \qquad \text{so} \qquad Ax = \sum_{j=1}^n \sum_{i=1}^m c_ja_{ij}\, y_i.$$
The matrix of A w.r.t. the two bases is $[A] = (a_{ij})$. (See Abstract Linear Algebra.)

2.3 Operator Norm

Transition: Euclidean space $\to$ Banach space.

Definition 2.5: Operator norm

If $A \in L(\mathbb{R}^n, \mathbb{R}^m)$, the *norm* of A is

$$\|A\| = \sup_{\|x\|=1} \|Ax\|,$$

where $\|\cdot\|$ is the Euclidean norm on $\mathbb{R}^m$. (Intuitively: how far A can stretch a unit vector—we pick the maximum.)

Theorem 2.6: Properties of the operator norm

1. $\|A\| = \sup_{\|x\|=1} \|Ax\|$ (definition).
2. $\|Ax\| \leq \|A\| \cdot \|x\|$ for all $x \in \mathbb{R}^n$.

Theorem 2.7: Operator norm is finite; A is uniformly continuous

If $A \in L(\mathbb{R}^n, \mathbb{R}^m)$, then $\|A\| < \infty$ and A is uniformly continuous. Moreover:

(b) $\|A + B\| \leq \|A\| + \|B\|$ and $\|cA\| = |c| \, \|A\|$.

(c) $L(\mathbb{R}^n, \mathbb{R}^m)$ with $d(A, B) = \|A - B\|$ is a metric space.

(d) If $A \in L(\mathbb{R}^n, \mathbb{R}^m)$ and $B \in L(\mathbb{R}^m, \mathbb{R}^k)$, then $\|BA\| \leq \|B\| \cdot \|A\|$.

Proof. **(1)** Let $x \in \mathbb{R}^n$ with $\|x\| = 1$, so $x = \sum_{j=1}^n c_j e_j$ in the standard basis. Then

$$\|Ax\| = \left\| \sum_{j=1}^n c_j A e_j \right\| \leq \sum_{j=1}^n |c_j| \, \|Ae_j\|.$$

Let $M = \max_j \|Ae_j\|$. By the Cauchy–Schwarz inequality for ℓ^1 vs ℓ^∞:

$$\|Ax\| \leq M \sum_{j=1}^n |c_j| \leq M\sqrt{n} < \infty.$$

Uniform continuity: Let $\varepsilon > 0$ and choose $\delta = \frac{\varepsilon}{\|A\|+1}$ (this δ is independent of x, so continuity is uniform). For $x, y \in \mathbb{R}^n$:

$$\|A(x - y)\| \leq \|A\| \cdot \|x - y\| < \|A\| \cdot \delta < \varepsilon.$$

(b) $\|A + B\| = \sup_{\|x\|=1} \|Ax + Bx\| \leq \sup \|Ax\| + \sup \|Bx\| = \|A\| + \|B\|$.
(c) HW.
(d) $\|BA\| = \sup_{\|x\|=1} \|B(Ax)\| \leq \sup_{\|x\|=1} \|B\| \, \|Ax\| = \|B\| \sup_{\|x\|=1} \|Ax\| = \|B\| \, \|A\|$. □

Theorem 2.8: 1-1 iff onto for $L(\mathbb{R}^n)$

$A \in L(\mathbb{R}^n)$ is injective $\iff$ A is surjective.

Remark.

> $A \in L(\mathbb{R}^n)$ is a bijection $\iff$ A is 1-1 $\iff$ A is onto. Also, A is 1-1 $\iff$ $Ax = 0 \Rightarrow x = 0$, i.e., $\ker(A) = \{0\}$.

Proof. **("Making the bijection proof easy.")** Let $B = \{x_1, \ldots, x_n\}$ be a basis for $\mathbb{R}^n$ and set $B' = \{Ax_1, \ldots, Ax_n\}$. Then $\operatorname{Im} A = \operatorname{span} B'$. Hence:

$$\begin{aligned} A \text{ is onto} &\iff \operatorname{Im} A = \mathbb{R}^n \iff \operatorname{span} B' = \mathbb{R}^n \iff B' \text{ is independent} \\ &\iff \left(\sum c_i A x_i = 0 \Rightarrow c_i = 0\right) \iff \left(A\left(\sum c_i x_i\right) = 0 \Rightarrow c_i = 0\right) \\ &\iff (Ax = 0 \Rightarrow x = 0) \iff A \text{ is 1-1.} \end{aligned}$$

□

2.4 Invertible Linear Maps: Open Set and Continuity

Theorem 2.9: Invertible maps form an open set

Let $\Omega = \{A \in L(\mathbb{R}^n) \mid A \text{ is invertible}\}$.

(1) If $A \in \Omega$ and $B \in L(\mathbb{R}^n)$ satisfy $\|A - B\| \cdot \|A^{-1}\| < 1$, then $B \in \Omega$.

(2) Ω is open in $L(\mathbb{R}^n)$.

(3) The map $T : \Omega \to \Omega$ defined by $T(A) = A^{-1}$ is continuous and bijective.

Proof. **(1)** It suffices to show B is 1-1 (then invoke the previous theorem). For any $x \in \mathbb{R}^n$:

$$\begin{aligned} \|x\| &= \left\|A^{-1}Ax\right\| \le \left\|A^{-1}\right\| \|Ax\| = \left\|A^{-1}\right\| \|(A - B + B)x\| \\ &\le \left\|A^{-1}\right\|(\|(A-B)x\| + \|Bx\|) \le \left\|A^{-1}\right\|(\|A - B\| \cdot \|x\| + \|Bx\|). \end{aligned}$$

Let $c = \|A^{-1}\| \|A - B\| < 1$. Rearranging:

$$(1 - c)\|x\| \le \left\|A^{-1}\right\| \|Bx\|. \tag{$*$}$$

If $Bx = 0$, then $\|x\| = 0$, i.e., $x = 0$. So B is 1-1.

(2) For $A \in \Omega$, take $r = \frac{1}{\|A^{-1}\|}$. Then $\|B - A\| < r \Rightarrow \|A - B\| \|A^{-1}\| < 1 \Rightarrow B \in \Omega$ by (1).

(3) *Goal:* show $B^{-1} \to A^{-1}$ as $B \to A$. Note:

$$B^{-1} - A^{-1} = B^{-1}(I - BA^{-1}) = B^{-1}(A - B)A^{-1},$$

so $\|B^{-1} - A^{-1}\| \leq \|B^{-1}\| \, \|A - B\| \, \|A^{-1}\|$.

First bound $\|B^{-1}\|$: set $\delta_0 = \frac{1}{2\|A^{-1}\|}$. If $\|B - A\| < \delta_0$, then $\|A - B\| \, \|A^{-1}\| < \frac{1}{2}$, and from $(*)$:

$$\|x\| \leq \frac{\|A^{-1}\|}{1-c} \|Bx\| < 2 \left\|A^{-1}\right\| \|Bx\| .$$

Applying this with $x = B^{-1}y$: $\|B^{-1}y\| \leq 2 \|A^{-1}\| \, \|y\|$, so $\|B^{-1}\| \leq 2 \|A^{-1}\|$.

Now let $\varepsilon > 0$; choose $\delta = \min\left\{\delta_0, \dfrac{\varepsilon}{2\|A^{-1}\|^2}\right\}$. Then for any $B \in B_\delta(A)$:

$$\left\|B^{-1} - A^{-1}\right\| \leq 2 \left\|A^{-1}\right\| \cdot \|B - A\| \cdot \left\|A^{-1}\right\| \leq 2 \left\|A^{-1}\right\|^2 \cdot \delta < \varepsilon.$$

□

2.5 Fréchet Derivative in $\mathbb{R}^n$

Definition 2.10: Differentiability in $\mathbb{R}^n$

Let $E \subseteq \mathbb{R}^n$ be open and $f : E \to \mathbb{R}^m$. Then f is *differentiable at* $x \in E$ if there exists $A \in L(\mathbb{R}^n, \mathbb{R}^m)$ such that

$$\lim_{h \to 0} \frac{\|f(x+h) - f(x) - Ah\|}{\|h\|} = 0.$$

A is called the *(total/Fréchet) derivative* of f at x, denoted $A = f'(x) = Df(x)$.

Remark.

1. This is the Fréchet derivative; it can be defined on any normed (Banach) space.
2. For $f : E \to \mathbb{R}^m$, the derivative $f'(x) \in L(\mathbb{R}^n, \mathbb{R}^m)$, and the map $f' : E \to L(\mathbb{R}^n, \mathbb{R}^m)$ is itself a function.
3. When $m = 1$: $f : E \to \mathbb{R}$, $f'(x) = Df(x) = \nabla f(x)$, the *gradient* of f at x.
4. Let $r(h) = f(x+h) - f(x) - f'(x)h$. Then $\frac{\|r(h)\|}{\|h\|} \to 0$ as $h \to 0$, so $f(x+h) \approx f(x) + f'(x)h$. Since $f'(x)h \to 0$ as $h \to 0$ (linearity), $f(x+h) \to f(x)$: differentiability implies continuity.

Example.

1. $f(x) = c$ (constant), $\forall x \in \mathbb{R}^n$. Then $Df(x) = 0$ (zero transformation).
2. $f(x) = Ax$ with $A \in L(\mathbb{R}^n, \mathbb{R}^m)$. We seek B with $\lim_{h\to 0} \frac{\|A(x+h)-Ax-Bh\|}{\|h\|} = 0$, i.e., $\frac{\|Ah-Bh\|}{\|h\|} \to 0$. Taking $B = A$ makes this 0. So $Df(x) = A$.
3. For $f : \mathbb{R} \to \mathbb{R}$: $Df(x)$ exists iff $\exists\, c \in \mathbb{R}$ with $\lim_{h\to 0} \frac{|f(x+h)-f(x)-ch|}{|h|} = 0$, which is equivalent to $\lim_{t\to x} \frac{f(t)-f(x)}{t-x} = c$. So the Fréchet and classical derivatives agree.

Theorem 2.11: Uniqueness of the derivative

If $f : E \to \mathbb{R}^m$ is differentiable at $x \in E$, then $f'(x)$ is unique.

Proof. Suppose $A, B \in L(\mathbb{R}^n, \mathbb{R}^m)$ both satisfy the differentiability condition. Then:

$$\begin{aligned}\frac{\|(A-B)h\|}{\|h\|} &= \frac{\|Ah - (f(x+h)-f(x)) + (f(x+h)-f(x)) - Bh\|}{\|h\|} \\ &\le \frac{\|f(x+h)-f(x)-Ah\|}{\|h\|} + \frac{\|f(x+h)-f(x)-Bh\|}{\|h\|} \to 0\end{aligned}$$

by assumption. Since $\frac{\|(A-B)h\|}{\|h\|} \to 0$ for all directions h, it follows that $A = B$. □

2.6 Partial Derivatives

Definition 2.12: Partial derivative

Let $E \subseteq \mathbb{R}^n$ be open, $f : E \to \mathbb{R}^m$, written $f = (f_1, \ldots, f_m)$ with each $f_i : E \to \mathbb{R}$, and $f(x) = \sum_{i=1}^m f_i(x)\, e_i$ (standard basis of $\mathbb{R}^m$).
The *partial derivative of f_i along direction e_j* is

$$D_j f_i(x) = \frac{\partial f_i}{\partial x_j}(x) = \lim_{t\to 0} \frac{f_i(x+te_j) - f_i(x)}{t}.$$

2.7 Chain Rule in $\mathbb{R}^n$

Theorem 2.13: Chain Rule

Let $E \subseteq \mathbb{R}^n$ and $E' \subseteq \mathbb{R}^m$ be open sets, $x_0 \in E$, $f : E \to \mathbb{R}^m$, $g : E' \to \mathbb{R}^k$, with $f(E) \subseteq E'$. Suppose f is differentiable at x_0 and g is differentiable at $f(x_0)$. Then $g \circ f$ is differentiable at x_0 and

$$(g \circ f)'(x_0) = g'(f(x_0)) \cdot f'(x_0).$$

(The right-hand side is a composition of linear transformations, i.e., matrix multiplication.)

Proof. Let $p = g \circ f$, $f'(x_0) = A$, $g'(f(x_0)) = B$. We want to show $\frac{\|p(x_0+h)-p(x_0)-BAh\|}{\|h\|} \to 0$ as $h \to 0$.

Since f is differentiable at x_0, define

$$r(h) = f(x_0 + h) - f(x_0) - Ah, \qquad \frac{\|r(h)\|}{\|h\|} \to 0 \text{ as } h \to 0.$$

Since g is differentiable at $f(x_0)$, define

$$s(k) = g(f(x_0) + k) - g(f(x_0)) - Bk, \qquad \frac{\|s(k)\|}{\|k\|} \to 0 \text{ as } k \to 0,\ s(0) = 0.$$

Now compute:

$$\begin{aligned}
p(x_0 + h) - p(x_0) - BAh &= g(f(x_0 + h)) - g(f(x_0)) - BAh \\
&= g\big(f(x_0 + h) - f(x_0) + f(x_0)\big) - g(f(x_0)) - BAh \\
&= B(f(x_0 + h) - f(x_0)) + s(f(x_0 + h) - f(x_0)) - BAh \\
&= B(Ah + r(h)) + s(Ah + r(h)) - BAh \\
&= B(r(h)) + s(Ah + r(h)).
\end{aligned}$$

As $h \to 0$: $\|Ah + r(h)\| \le \|A\|\,\|h\| + \|r(h)\| \to 0$, so the argument of s goes to 0, and

$$\frac{\|B(r(h))\|}{\|h\|} \le \|B\| \frac{\|r(h)\|}{\|h\|} \to 0, \qquad \frac{\|s(Ah + r(h))\|}{\|h\|} \to 0$$

(the latter because $\frac{\|s(k)\|}{\|k\|} \to 0$ as $k \to 0$ and $\frac{\|Ah+r(h)\|}{\|h\|} \to \|A\| < \infty$). Hence $\frac{\|p(x_0+h)-p(x_0)-BAh\|}{\|h\|} \to 0$. $\square$

2.8 Directional Derivatives and the Total Derivative

Definition 2.14: Directional derivative

Let $E \subseteq \mathbb{R}^n$ be open, $f : E \to \mathbb{R}^m$, $x \in E$, and $u \in \mathbb{R}^n$ with $\|u\| = 1$. The *directional derivative of f at x in the direction u* is

$$D_u f(x) = \lim_{t \to 0} \frac{f(x + tu) - f(x)}{t},$$

provided the limit exists.

Remark.

When $u = e_j$ (the j-th standard basis vector), $D_{e_j} f_i(x) = D_j f_i(x)$, the partial derivative.

Theorem 2.15: Total derivative vs. partial derivatives

Let $E \subseteq \mathbb{R}^n$ be open and $f : E \to \mathbb{R}^m$. If f is differentiable at x, then all partial derivatives $D_j f_i(x)$ exist, and

$$f'(x)e_j = D_j f(x) \qquad (j = 1, \ldots, n).$$

In other words, the matrix of $f'(x)$ (the *Jacobian matrix*) is

$$[f'(x)] = \begin{pmatrix} D_1 f_1(x) & \cdots & D_n f_1(x) \\ \vdots & & \vdots \\ D_1 f_m(x) & \cdots & D_n f_m(x) \end{pmatrix}.$$

Proof. Fix j and let $h = te_j$. Then $\|h\| = |t|$. By differentiability:

$$\lim_{t \to 0} \frac{\|f(x + te_j) - f(x) - f'(x)(te_j)\|}{|t|} = 0,$$

which means $\lim_{t \to 0} \frac{f(x+te_j)-f(x)}{t} = f'(x)e_j$. Looking at the i-th component gives $D_j f_i(x) = [f'(x)]_{ij}$. $\square$

Remark.

The converse is **false**: partial derivatives can exist without f being differentiable. However, continuity of all partials *does* imply differentiability (see C^1 theorem below).

Definition 2.16: $C^1(E)$

Let $E \subseteq \mathbb{R}^n$ be open and $f : E \to \mathbb{R}^m$. We say $f \in C^1(E)$ (or f is *continuously differentiable on E*) if all partial derivatives $D_j f_i$ exist on E and are continuous on E.

Theorem 2.17: $C^1 \iff$ differentiable with continuous derivative

Let $E \subseteq \mathbb{R}^n$ be open. Then $f \in C^1(E)$ if and only if f is differentiable on E and $f' : E \to L(\mathbb{R}^n, \mathbb{R}^m)$ is continuous.

Proof. We prove the harder direction: if all $D_j f_i$ exist and are continuous, then f is differentiable.

It suffices to treat $m = 1$ (the general case follows component-wise). Fix $x \in E$ and let $h = (h_1, \ldots, h_n)$ be small. Write

$$f(x+h) - f(x) = \sum_{j=1}^{n} \big[f(x+u_j) - f(x+u_{j-1})\big],$$

where $u_0 = 0$, $u_j = (h_1, \ldots, h_j, 0, \ldots, 0)$. Each term changes only the j-th coordinate by h_j, so by the (1-dimensional) MVT, there exists θ_j between 0 and h_j such that

$$f(x+u_j) - f(x+u_{j-1}) = D_j f(x + u_{j-1} + \theta_j e_j)\, h_j.$$

Therefore

$$f(x+h) - f(x) - \sum_{j=1}^{n} D_j f(x)\, h_j = \sum_{j=1}^{n} \big[D_j f(x + u_{j-1} + \theta_j e_j) - D_j f(x)\big] h_j.$$

Dividing by $\|h\|$ and using $|h_j| \le \|h\|$:

$$\left\| \frac{f(x+h) - f(x) - Ah}{\|h\|} \right\| \le \sum_{j=1}^{n} \big| D_j f(x + u_{j-1} + \theta_j e_j) - D_j f(x) \big| \ \to 0$$

as $h \to 0$, since each $D_j f$ is continuous at x. Hence $f'(x)h = \sum_j D_j f(x)\, h_j$. □

2.9 Mean Value Theorem in $\mathbb{R}^n$

Remark.

> The usual MVT fails for $f : E \to \mathbb{R}^m$ with $m > 1$ (see Remark **??**). However, a *norm* version holds.

Theorem 2.18: MVT in $\mathbb{R}^n$

Let $E \subseteq \mathbb{R}^n$ be open and convex, and let $f : E \to \mathbb{R}^m$ be differentiable on E. For any $x, y \in E$ (with $[x, y] \subseteq E$):

$$\|f(y) - f(x)\| \le \|y - x\| \cdot \sup_{t \in [0,1]} \big\| f'\big(x + t(y-x)\big) \big\|.$$

Proof. Let $\phi : [0,1] \to \mathbb{R}^m$ be $\phi(t) = f(x+t(y-x))$. By the chain rule, $\phi'(t) = f'(x+t(y-x)) \cdot (y-x)$. Since ϕ is vector-valued, we cannot apply the 1-D MVT directly. Instead, fix any $u \in \mathbb{R}^m$ with $\|u\| = 1$ and apply the 1-D MVT to $g(t) = \langle u, \phi(t) \rangle$:

$$g(1) - g(0) = g'(\lambda) = \langle u, \phi'(\lambda) \rangle \leq \|u\| \, \|\phi'(\lambda)\| = \|\phi'(\lambda)\|$$

for some $\lambda \in (0,1)$. Taking the supremum over u gives $\|\phi(1) - \phi(0)\| \leq \sup_{t \in [0,1]} \|\phi'(t)\|$. Substituting back yields the result. □

2.10 Contraction Mapping Theorem

Definition 2.19: Contraction

Let (X, d) be a complete metric space. A map $\phi : X \to X$ is a *contraction* if there exists $c \in (0,1)$ such that

$$d(\phi(x), \phi(y)) \leq c\, d(x,y) \qquad \forall x, y \in X.$$

Theorem 2.20: Contraction Mapping / Banach Fixed-Point Theorem

If $\phi : X \to X$ is a contraction on a complete metric space (X, d), then ϕ has a **unique fixed point**: there exists exactly one $x^* \in X$ with $\phi(x^*) = x^*$.

Proof. **Existence.** Pick any $x_0 \in X$ and define $x_{k+1} = \phi(x_k)$. Then for $k \geq 1$:

$$d(x_{k+1}, x_k) = d(\phi(x_k), \phi(x_{k-1})) \leq c\, d(x_k, x_{k-1}) \leq \cdots \leq c^k d(x_1, x_0).$$

For $m > n$:

$$d(x_m, x_n) \leq \sum_{k=n}^{m-1} d(x_{k+1}, x_k) \leq d(x_1, x_0) \sum_{k=n}^{\infty} c^k = \frac{c^n}{1-c} d(x_1, x_0) \to 0.$$

So (x_k) is Cauchy, hence converges to some $x^* \in X$ (completeness). Since ϕ is continuous (contractions are Lipschitz):

$$x^* = \lim_k x_{k+1} = \lim_k \phi(x_k) = \phi(x^*).$$

Uniqueness. If $\phi(y) = y$ as well, then $d(x^*, y) = d(\phi(x^*), \phi(y)) \leq c\, d(x^*, y)$, so $(1-c)d(x^*, y) \leq 0$, giving $d(x^*, y) = 0$, i.e., $x^* = y$. □

2.11 Inverse Function Theorem

Theorem 2.21: Inverse Function Theorem (IFT)

Let $E \subseteq \mathbb{R}^n$ be open, $f \in C^1(E, \mathbb{R}^n)$, and $a \in E$. Suppose $f'(a)$ is invertible (i.e., $f'(a) \in \Omega$). Then:

1. There exists an open set $U \ni a$ and an open set $V \ni f(a)$ such that $f : U \to V$ is a bijection.
2. The inverse $g = f^{-1} : V \to U$ is C^1, and

$$g'(y) = \left[f'(g(y))\right]^{-1} \qquad \forall y \in V.$$

Proof. **Setup.** Let $A = f'(a)$ (invertible). Define $\phi(x) = x - A^{-1}(f(x) - y)$ for fixed y near $f(a)$. Finding $f(x) = y$ is equivalent to finding a fixed point of ϕ.

Step 1: ϕ is a contraction near a. Since $f \in C^1$, f' is continuous at a. Choose $r > 0$ small enough that for all $x \in B = \overline{B(a, r)}$:

$$\|f'(x) - A\| \leq \frac{1}{2\left\|A^{-1}\right\|}.$$

Then

$$\phi'(x) = I - A^{-1}f'(x) = A^{-1}(A - f'(x)),$$

so

$$\|\phi'(x)\| \leq \left\|A^{-1}\right\| \|A - f'(x)\| \leq \left\|A^{-1}\right\| \cdot \frac{1}{2\left\|A^{-1}\right\|} = \tfrac{1}{2}.$$

By the MVT (for $\mathbb{R}^n$): $\|\phi(x_1) - \phi(x_2)\| \leq \frac{1}{2}\|x_1 - x_2\|$ for $x_1, x_2 \in B$. Hence ϕ is a contraction on B.

Step 2: ϕ maps B to itself (for y near $f(a)$). For $x \in B$:

$$\|\phi(x) - a\| \leq \|\phi(x) - \phi(a)\| + \|\phi(a) - a\| \leq \tfrac{1}{2}\|x - a\| + \left\|A^{-1}\right\| \|f(a) - y\|.$$

If $\|y - f(a)\| \leq \dfrac{r}{2\left\|A^{-1}\right\|}$, then $\|\phi(x) - a\| \leq r$, so $\phi(x) \in B$.

Step 3: Fixed point $\Rightarrow$ local bijection. By the CMT, ϕ has a unique fixed point $x \in B$, giving the unique solution to $f(x) = y$. Setting $V = \{y : \|y - f(a)\| < \frac{r}{2\|A^{-1}\|}\}$ and $U = f^{-1}(V) \cap B$ gives the desired open sets.

Step 4: $g \in C^1$. Since $f'(g(y)) \in \Omega$ (by continuity and the fact that g stays near a), the formula $g'(y) = [f'(g(y))]^{-1}$ follows from differentiating $f(g(y)) = y$. Continuity of g' follows from continuity of f', g, and the inversion map $A \mapsto A^{-1}$. $\square$

2.12 Implicit Function Theorem

Theorem 2.22: Implicit Function Theorem

Let $E \subseteq \mathbb{R}^{n+m}$ be open and $F : E \to \mathbb{R}^m$ be in $C^1(E)$. Write points in $\mathbb{R}^{n+m}$ as (x, y) with $x \in \mathbb{R}^n$, $y \in \mathbb{R}^m$. Suppose $(a, b) \in E$ satisfies $F(a, b) = 0$ and the $m \times m$ matrix

$$A_y = \big(D_{n+j}F_i(a, b)\big)_{1\le i,j\le m}$$

(the Jacobian of F with respect to the y-variables) is invertible.
Then there exist open sets $U \subseteq \mathbb{R}^n$ with $a \in U$ and $W \subseteq \mathbb{R}^{n+m}$ with $(a, b) \in W$ such that:

1. For every $x \in U$ there exists a **unique** $y = g(x) \in \mathbb{R}^m$ with $(x, g(x)) \in W$ and $F(x, g(x)) = 0$.
2. The resulting $g : U \to \mathbb{R}^m$ is in $C^1(U)$, with

$$g'(x) = -\big[A_y(x, g(x))\big]^{-1} \cdot A_x(x, g(x)),$$

where $A_x = \big(D_jF_i\big)_{1\le i\le m,\, 1\le j\le n}$ is the Jacobian of F with respect to x.

Proof. **Reduction to IFT.** Define $\Phi : E \to \mathbb{R}^{n+m}$ by $\Phi(x, y) = (x, F(x, y))$. Then

$$\Phi'(a, b) = \begin{pmatrix} I_n & 0 \\ A_x & A_y \end{pmatrix},$$

which is invertible $\iff$ A_y is invertible (by block-matrix determinant).

By the IFT applied to Φ at (a, b), there exist open neighborhoods W of (a, b) and $\tilde{V}$ of $\Phi(a, b) = (a, 0)$ such that $\Phi : W \to \tilde{V}$ is a C^1 bijection. Define g by the implicit equation: given $x \in U$ (a neighborhood of a), there is a unique $y = g(x)$ with $(x, y) \in W$ and $\Phi(x, y) = (x, 0)$, i.e., $F(x, y) = 0$.

The formula for g' follows by differentiating $F(x, g(x)) = 0$ using the chain rule:

$$A_x + A_y\, g'(x) = 0 \quad \Rightarrow \quad g'(x) = -A_y^{-1}A_x.$$

□

Example.

Let $F(x, y) = x^2 + y^2 - 1$. Then $F(1, 0) = 0$ and $\frac{\partial F}{\partial y}(1, 0) = 2y\big|_{(1,0)} = 0$. Since $\frac{\partial F}{\partial y} = 0$ at $(1, 0)$, the IFT does **not** apply there.

However at $(0, 1)$: $\frac{\partial F}{\partial y}(0, 1) = 2 \neq 0$, so IFT applies: near $x = 0$, the circle is the graph of a C^1 function $y = g(x) = \sqrt{1 - x^2}$. Moreover,

$$g'(x) = -\left(\frac{\partial F}{\partial y}\right)^{-1}\frac{\partial F}{\partial x} = -\frac{1}{2y} \cdot 2x = -\frac{x}{y} = -\frac{x}{\sqrt{1 - x^2}}.$$

3 Riemann–Stieltjes Integration

3.1 Setup

Throughout this section, let $\alpha : [a, b] \to \mathbb{R}$ be a non-decreasing function (called the *integrator*) and $f : [a, b] \to \mathbb{R}$ be bounded.

Definition 3.1: Partition and Darboux sums

1. A *partition* P of $[a, b]$ is a finite set $\{x_0, x_1, \ldots, x_N\}$ with $a = x_0 < x_1 < \cdots < x_N = b$.
2. Write $\Delta\alpha_i = \alpha(x_i) - \alpha(x_{i-1}) \geq 0$ for $i = 1, \ldots, N$.
3. The *upper* and *lower Darboux sums* are

$$U(f, P, \alpha) = \sum_{i=1}^{N} M_i \, \Delta\alpha_i, \qquad M_i = \sup_{x \in [x_{i-1}, x_i]} f(x),$$
$$L(f, P, \alpha) = \sum_{i=1}^{N} m_i \, \Delta\alpha_i, \qquad m_i = \inf_{x \in [x_{i-1}, x_i]} f(x).$$

4. The *upper* and *lower Riemann–Stieltjes integrals* are

$$\overline{\int_a^b} f \, d\alpha = \inf_P U(f, P, \alpha), \qquad \underline{\int_a^b} f \, d\alpha = \sup_P L(f, P, \alpha).$$

Remark.

> Always $L(f, P, \alpha) \leq U(f, P, \alpha)$ and $\underline{\int} f \, d\alpha \leq \overline{\int} f \, d\alpha$.

Definition 3.2: Riemann–Stieltjes integral, class $\mathscr{R}(\alpha)$

f is *Riemann–Stieltjes integrable with respect to* α on $[a, b]$, written $f \in \mathscr{R}(\alpha)$, if

$$\overline{\int_a^b} f \, d\alpha = \underline{\int_a^b} f \, d\alpha.$$

The common value is the *Riemann–Stieltjes integral*, denoted $\displaystyle\int_a^b f \, d\alpha$.

3.2 Refinements and Integrability Criterion

Definition 3.3: Refinement

P^* is a *refinement* of P if $P \subseteq P^*$ (as sets of points).

Theorem 3.4: Refinement monotonicity

If P^* is a refinement of P, then

$$L(f, P, \alpha) \leq L(f, P^*, \alpha) \leq U(f, P^*, \alpha) \leq U(f, P, \alpha).$$

Theorem 3.5: Integrability criterion (ε-criterion)

$f \in \mathscr{R}(\alpha)$ if and only if for every $\varepsilon > 0$ there exists a partition P such that

$$U(f, P, \alpha) - L(f, P, \alpha) < \varepsilon.$$

3.3 Sufficient Conditions for Integrability

Theorem 3.6: Continuous $\Rightarrow$ integrable

If $f : [a, b] \to \mathbb{R}$ is continuous and α is non-decreasing, then $f \in \mathscr{R}(\alpha)$.

Proof. Since $[a, b]$ is compact, f is uniformly continuous. Given $\varepsilon > 0$, choose $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ whenever $|x - y| < \delta$. Pick a partition P with all subintervals of length less than δ. Then for each i: $M_i - m_i < \varepsilon$. Hence

$$U(f, P, \alpha) - L(f, P, \alpha) = \sum_i (M_i - m_i)\Delta\alpha_i < \varepsilon \sum_i \Delta\alpha_i = \varepsilon\big(\alpha(b) - \alpha(a)\big).$$

Since ε is arbitrary, the criterion holds. $\square$

Theorem 3.7: Monotone f + continuous α $\Rightarrow$ integrable

If $f : [a, b] \to \mathbb{R}$ is monotone (say non-decreasing) and α is continuous on $[a, b]$, then $f \in \mathscr{R}(\alpha)$.

Proof. Given $\varepsilon > 0$, choose $n \in \mathbb{N}$ such that $[f(b) - f(a)]\frac{\alpha(b)-\alpha(a)}{n} < \varepsilon$. Partition so that $\Delta\alpha_i = \frac{\alpha(b)-\alpha(a)}{n}$ for all i (possible by IVT for α, since α is continuous). Since f is non-decreasing, $M_i - m_i = f(x_i) - f(x_{i-1})$. Then

$$U - L = \sum_i (M_i - m_i)\Delta\alpha_i = \frac{\alpha(b) - \alpha(a)}{n} \sum_i \big(f(x_i) - f(x_{i-1})\big) = \frac{\alpha(b) - \alpha(a)}{n}\big(f(b) - f(a)\big) < \varepsilon.$$

$\square$

3.4 Properties of the Riemann–Stieltjes Integral

Theorem 3.8: Properties

Let $f, g \in \mathscr{R}(\alpha)$ on $[a, b]$.

1. **Linearity:** $\int (cf + dg)\, d\alpha = c \int f\, d\alpha + d \int g\, d\alpha$ for $c, d \in \mathbb{R}$.
2. **Additivity in integrator:** If α, β are non-decreasing and $c > 0$, then $f \in \mathscr{R}(\alpha + \beta)$ and $\int f\, d(\alpha + \beta) = \int f\, d\alpha + \int f\, d\beta$; also $f \in \mathscr{R}(c\alpha)$ and $\int f\, d(c\alpha) = c \int f\, d\alpha$.
3. **Additivity in interval:** If $a < c < b$ and $f \in \mathscr{R}(\alpha)$ on $[a, b]$, then $f \in \mathscr{R}(\alpha)$ on $[a, c]$ and $[c, b]$, and $\int_a^b f\, d\alpha = \int_a^c f\, d\alpha + \int_c^b f\, d\alpha$.
4. **Monotonicity:** If $f \leq g$ on $[a, b]$, then $\int f\, d\alpha \leq \int g\, d\alpha$.
5. **Bound:** $\left|\int f\, d\alpha\right| \leq \sup_{[a,b]} |f| \cdot (\alpha(b) - \alpha(a))$.
6. **Absolute value:** $|f| \in \mathscr{R}(\alpha)$ and $\left|\int f\, d\alpha\right| \leq \int |f|\, d\alpha$.
7. **Product:** If $f, g \in \mathscr{R}(\alpha)$, then $fg \in \mathscr{R}(\alpha)$.
8. **Composition:** If $f \in \mathscr{R}(\alpha)$, $\phi : [m, M] \to \mathbb{R}$ is continuous, and $m \leq f \leq M$, then $\phi \circ f \in \mathscr{R}(\alpha)$.

3.5 Integration and Differentiation

Theorem 3.9: Differentiable integrator

Suppose α is differentiable on $[a, b]$ with $\alpha' \in \mathscr{R}$ (the ordinary Riemann integral). If $f : [a, b] \to \mathbb{R}$ is bounded, then

$$f \in \mathscr{R}(\alpha) \iff f\alpha' \in \mathscr{R},$$

and in that case $\displaystyle\int_a^b f\, d\alpha = \int_a^b f(x)\, \alpha'(x)\, dx$.

Theorem 3.10: Change of variables

Suppose $\phi : [c, d] \to [a, b]$ is a strictly increasing, C^1 surjection, $\alpha : [a, b] \to \mathbb{R}$ is non-decreasing, and $f \in \mathscr{R}(\alpha)$. Define $\beta = \alpha \circ \phi$ and $g = f \circ \phi$. Then $g \in \mathscr{R}(\beta)$ and

$$\int_c^d g \, d\beta = \int_a^b f \, d\alpha.$$

4 Sequences and Series of Functions

4.1 Pointwise and Uniform Convergence

Definition 4.1: Pointwise convergence

Let E be a set, $f_n : E \to \mathbb{R}$ (or $\to \mathbb{C}$), and $f : E \to \mathbb{R}$. We say f_n *converges pointwise to* f *on* E if for every $x \in E$:

$$f_n(x) \to f(x) \quad \text{as } n \to \infty.$$

(Equivalently, $\forall x \in E$, $\forall \varepsilon > 0$, $\exists N = N(\varepsilon, x)$ such that $n \geq N \Rightarrow |f_n(x) - f(x)| < \varepsilon$.)
For a series $\sum g_n$: it converges pointwise if the partial sums $f_n = \sum_{k=1}^n g_k$ converge pointwise.

Definition 4.2: Uniform convergence

f_n *converges uniformly to* f *on* E if

$$\sup_{x \in E} |f_n(x) - f(x)| \to 0 \quad \text{as } n \to \infty.$$

(Equivalently, $\forall \varepsilon > 0$, $\exists N = N(\varepsilon)$ (independent of x) such that $n \geq N \Rightarrow |f_n(x) - f(x)| < \varepsilon$ for all $x \in E$.)
For a series: $\sum g_n$ converges uniformly if its partial sums do.

Remark.

Uniform convergence $\Rightarrow$ pointwise convergence, but not conversely.
Example: $f_n(x) = x^n$ on $[0, 1]$. $f_n \to f$ pointwise where $f(x) = 0$ for $x \in [0, 1)$ and $f(1) = 1$. But $\sup_{x \in [0,1]} |f_n(x) - f(x)| = 1 \not\to 0$, so convergence is not uniform.

Remark.

What pointwise convergence does NOT preserve:

1. Continuity: the pointwise limit of continuous functions may be discontinuous (e.g., x^n on $[0,1]$).
2. Integrability: $\lim \int f_n \neq \int \lim f_n$ in general.
3. Differentiability: $\lim f_n'$ need not equal $(\lim f_n)'$.

Uniform convergence **does** preserve continuity, integrability, and (with extra hypotheses) differentiability.

4.2 Cauchy Criterion for Uniform Convergence

Theorem 4.3: Cauchy criterion (uniform convergence)

f_n converges uniformly on E if and only if:

$$\forall \varepsilon > 0,\ \exists N \text{ such that } m, n \geq N \Rightarrow \sup_{x \in E} |f_m(x) - f_n(x)| < \varepsilon.$$

Proof. ($\Rightarrow$) If $f_n \to f$ uniformly, then for $m, n \geq N$: $|f_m(x) - f_n(x)| \leq |f_m(x) - f(x)| + |f(x) - f_n(x)| < \varepsilon$.

($\Leftarrow$) The Cauchy condition means each sequence $(f_n(x))$ is Cauchy in $\mathbb{R}$, hence converges to some $f(x)$. Given $\varepsilon > 0$, choose N from the Cauchy condition. For $n \geq N$ and any x: let $m \to \infty$ to get $|f_n(x) - f(x)| \leq \varepsilon$. So convergence is uniform. $\square$

4.3 Weierstrass M-Test

Theorem 4.4: Weierstrass M-Test

Suppose $|f_n(x)| \leq M_n$ for all $x \in E$ and all n, and $\sum_{n=1}^{\infty} M_n < \infty$. Then $\sum f_n$ converges uniformly and absolutely on E.

Proof. Let $S_n = \sum_{k=1}^{n} f_k$. For $m > n$:

$$\sup_{x \in E} |S_m(x) - S_n(x)| \leq \sup_{x \in E} \sum_{k=n+1}^{m} |f_k(x)| \leq \sum_{k=n+1}^{m} M_k \to 0$$

as $n \to \infty$ (since $\sum M_k$ converges). By the Cauchy criterion, S_n converges uniformly. $\square$

4.4 Uniform Convergence and Continuity

Theorem 4.5: Interchange of limits

Let E be a metric space, $f_n : E \to \mathbb{R}$, and $p \in E$ a limit point. Suppose:

- $f_n \to f$ uniformly on E, and
- $\lim\limits_{x \to p} f_n(x) = a_n$ exists for each n.

Then $\lim_{n\to\infty} a_n$ exists and

$$\lim_{x\to p} f(x) = \lim_{n\to\infty} a_n = \lim_{n\to\infty} \lim_{x\to p} f_n(x).$$

In other words, $\lim\limits_{x\to p} \lim\limits_{n\to\infty} f_n(x) = \lim\limits_{n\to\infty} \lim\limits_{x\to p} f_n(x)$.

Proof. We show (a_n) is Cauchy. By uniform convergence, given $\varepsilon > 0$ choose N such that $m, n \geq N$ implies $\sup |f_m - f_n| < \varepsilon$. Then for $x \neq p$ and all $n, m \geq N$:

$$|a_m - a_n| \leq |a_m - f_m(x)| + |f_m(x) - f_n(x)| + |f_n(x) - a_n|.$$

Take $x \to p$ and use the limits: $|a_m - a_n| \leq \varepsilon$. So $(a_n) \to a$ (say).

Now show $\lim_{x\to p} f(x) = a$. For x near p and $n \geq N$:

$$|f(x) - a| \leq |f(x) - f_n(x)| + |f_n(x) - a_n| + |a_n - a| < \varepsilon + |f_n(x) - a_n| + \varepsilon.$$

Choose δ so $|x - p| < \delta$ implies $|f_n(x) - a_n| < \varepsilon$. Done. $\square$

Corollary 4.6

If $f_n : E \to \mathbb{R}$ are continuous and $f_n \to f$ uniformly on E, then f is continuous on E.

4.5 Dini's Theorem

Theorem 4.7: Dini's Theorem

Let K be a compact metric space, $f_n : K \to \mathbb{R}$ continuous, and $f : K \to \mathbb{R}$ continuous. Suppose $f_n \to f$ pointwise on K and the convergence is **monotone**: for every $x \in K$, $f_n(x)$ is non-decreasing (or non-increasing) in n. Then $f_n \to f$ uniformly on K.

Proof. WLOG $f_n \searrow f$ (non-increasing). Let $g_n = f_n - f \geq 0$; then $g_n \to 0$ pointwise, g_n is continuous, $g_n \searrow 0$.

Given $\varepsilon > 0$, let $K_n = \{x \in K : g_n(x) \geq \varepsilon\}$. This is a closed (hence compact) subset of K. Since $g_n \searrow 0$: $K_1 \supseteq K_2 \supseteq \cdots$ and $\bigcap_n K_n = \emptyset$ (for every x, $g_n(x) < \varepsilon$ for large n).

By the finite intersection property for compact sets, some $K_N = \emptyset$. This means $g_N(x) < \varepsilon$ for all $x \in K$. Since $g_n \searrow$: $g_n(x) \leq g_N(x) < \varepsilon$ for all $n \geq N$ and all x. Hence $\sup_{x \in K} g_n(x) < \varepsilon$ for $n \geq N$. $\square$

4.6 Integration and Uniform Convergence

Theorem 4.8: Uniform convergence and integration

Let α be non-decreasing on $[a, b]$, $f_n \in \mathscr{R}(\alpha)$, and $f_n \to f$ uniformly on $[a, b]$. Then $f \in \mathscr{R}(\alpha)$ and

$$\int_a^b f \, d\alpha = \lim_{n \to \infty} \int_a^b f_n \, d\alpha.$$

Proof. Let $\varepsilon_n = \sup_x |f_n(x) - f(x)| \to 0$. Then $f_n - \varepsilon_n \leq f \leq f_n + \varepsilon_n$, so

$$L(f_n, P, \alpha) - \varepsilon_n(\alpha(b) - \alpha(a)) \leq L(f, P, \alpha) \leq U(f, P, \alpha) \leq U(f_n, P, \alpha) + \varepsilon_n(\alpha(b) - \alpha(a)).$$

For any $\varepsilon > 0$ and n large enough ($\varepsilon_n < \varepsilon$), choose P so $U(f_n, P, \alpha) - L(f_n, P, \alpha) < \varepsilon$. Then $U(f, P, \alpha) - L(f, P, \alpha) < \varepsilon + 2\varepsilon_n(\alpha(b) - \alpha(a)) \to 0$. So $f \in \mathscr{R}(\alpha)$.

Moreover,

$$\left| \int f \, d\alpha - \int f_n \, d\alpha \right| \leq \int |f - f_n| \, d\alpha \leq \varepsilon_n(\alpha(b) - \alpha(a)) \to 0.$$

$\square$

4.7 Differentiation and Uniform Convergence

Theorem 4.9: Uniform convergence and differentiation

Let $f_n : [a, b] \to \mathbb{R}$ be differentiable with $f_n' \in \mathscr{R}$. Suppose:

- $f_n(x_0)$ converges for some $x_0 \in [a, b]$, and
- f_n' converges **uniformly** on $[a, b]$ to some function g.

Then f_n converges uniformly to a differentiable function f, and $f' = g$.

Proof. **Step 1: $f_n \to f$ uniformly.** For $x \in [a, b]$, write $f_n(x) - f_m(x) = [f_n(x) - f_m(x) - f_n(x_0) + f_m(x_0)] + [f_n(x_0) - f_m(x_0)]$. By the MVT applied to $f_n - f_m$:

$$|f_n(x) - f_m(x) - f_n(x_0) + f_m(x_0)| \leq (b - a) \sup |f_n' - f_m'|.$$

Since f_n' is uniformly Cauchy and $f_n(x_0)$ is Cauchy, (f_n) is uniformly Cauchy, hence uniformly convergent to some f.

Step 2: $f' = g$**.** Fix $x \in [a, b]$ and define, for $t \neq x$:

$$\phi_n(t) = \frac{f_n(t) - f_n(x)}{t - x}, \quad \phi(t) = \frac{f(t) - f(x)}{t - x}.$$

Since $f_n \to f$ uniformly, $\phi_n \to \phi$ uniformly in t (for t bounded away from x). Since $\lim_{t \to x} \phi_n(t) = f_n'(x) \to g(x)$ uniformly in n (as $f_n' \rightrightarrows g$), by the interchange-of-limits theorem:

$$f'(x) = \lim_{t \to x} \phi(t) = \lim_{t \to x} \lim_{n} \phi_n(t) = \lim_{n} \lim_{t \to x} \phi_n(t) = g(x).$$

□

Remark.

> **Weierstrass function.** The function
>
> $$f(x) = \sum_{n=0}^{\infty} a^n \cos(b^n \pi x), \quad 0 < a < 1,\ b \text{ a positive odd integer},\ ab > 1 + \tfrac{3\pi}{2},$$
>
> is continuous everywhere but differentiable nowhere. (The series converges uniformly by the M-test; the derivative fails to exist at every point.)

4.8 Equicontinuity

Definition 4.10: Equicontinuity

Let $\{f_n\}$ be a sequence of functions on a metric space E.

1. $\{f_n\}$ is *pointwise bounded on* E if $\sup_n |f_n(x)| < \infty$ for every fixed $x \in E$.
2. $\{f_n\}$ is *uniformly bounded on* E if $\sup_n \sup_{x \in E} |f_n(x)| < \infty$.
3. $\{f_n\}$ is *equicontinuous at* x_0 if for every $\varepsilon > 0$, there exists $\delta > 0$ such that for all n and all x with $d(x, x_0) < \delta$: $|f_n(x) - f_n(x_0)| < \varepsilon$. (The same δ works for all n.)
4. $\{f_n\}$ is *pointwise equicontinuous on* E if it is equicontinuous at every $x_0 \in E$.
5. $\{f_n\}$ is *uniformly equicontinuous on* E if for every $\varepsilon > 0$, there exists $\delta > 0$ such that for all n and all $x, y \in E$ with $d(x, y) < \delta$: $|f_n(x) - f_n(y)| < \varepsilon$.

Remark.

> Equicontinuity is a uniform version of continuity across the whole family $\{f_n\}$. A single continuous function on a compact set is uniformly continuous; similarly, an equicontinuous family on a compact set is *uniformly* equicontinuous.